EXPERIMENT - 7

Integration Using a Zoho Connection and API in Zoho Creator

Library Book Issue Management

Application Name	Library Book Issue Management
Form Name	Book_Issue_Request_Form
Report Name	Book_Issue_Request_Report
Student Name	CALEB D
Student Roll No.	241401016
Experiment	7
Platform	Zoho Creator
Record Date	09 October 2026

Lab Record	Authorized integration with Zoho Mail through a named Connection and API, with a structured HTML confirmation and evidence of delivery to the approved test mailbox.

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1. Objective

To integrate the Creator application with Zoho Mail through a proper authorized Connection/API and send a confirmation when a book request becomes available.

2. Integration Architecture

Staff marks Available > Notify_Available_Request > Send_Availability_Confirmation(request_id) > library_mail_connection > Zoho Mail API > member email.

3. Connection Configuration

Setting	Configured value
Service	Zoho Mail, India data center
Connection display name	Library_Mail_Connection
Connection link name	library_mail_connection
Permission	Send Email scope
Application access	Enabled for Library Book Issue Management
Sender mailbox	caleb1510@zohomail.in
Authentication	OAuth authorization completed by the account owner
Function	Send_Availability_Confirmation(int request_id)

The initial account had no Mail mailbox. The owner created a personal Zoho Mail mailbox and authorized the connection. No account password or token is embedded in this lab record.

4. API Behavior

The function retrieves the request, builds the subject and HTML body, then sends through the Zoho Mail account messages API using the named Connection. The request Email_Address is the recipient; the sender is the authorized Zoho mailbox.

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5. Implementation Procedure

1. Confirmed that the owner had a usable Zoho Mail mailbox in the India region.

2. Created and authorized Library_Mail_Connection with the Send Email permission.

3. Enabled the connection for the library application.

4. Implemented Send_Availability_Confirmation to retrieve the request and assemble the message.

5. Connected Notify_Available_Request to the function for Available records with Confirmation_Sent not true.

6. Ran the approved fictitious confirmation test to the owner mailbox and inspected the received message.

7. Refined the email into a greeting, details table and collection instructions, and verified the revised rendering.

6. Function Processing Logic

Step	Action
1. Retrieve	Find Book_Issue_Request_Form record by request_id.
2. Guard	Skip mock recipients ending @example.com.
3. Compose	Create request-specific subject and structured HTML body.
4. Send	Call Zoho Mail through library_mail_connection.
5. Confirm	Set Confirmation_Sent true only when the returned success code is 200.
6. Avoid repeat	The workflow skips ordinary edits when Confirmation_Sent is already true.

A successful API response indicates acceptance by the service. The received-message screenshot separately demonstrates actual delivery for the approved test mailbox; it does not guarantee delivery to every future recipient.

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6.1 Deluge API Call and Response Handling

The following excerpt was verified from the saved function in Creator. The HTML message is constructed earlier in the function and placed in the payload. The numeric account identifier selects the authorized sender mailbox; it is not an OAuth credential.

```
payload.put("content",message_body);
payload.put("mailFormat","html");
try
{
  response = invokeurl
  [
    url :"https://mail.zoho.in/api/accounts/7258794000000002002/messages"
    type :POST
    parameters:payload.toString()
    headers:{"Content-Type":"application/json"}
    connection:"library_mail_connection"
  ];
  if(response.get("status") != null &&
     response.get("status").get("code") == 200)
  {
    request.Confirmation_Sent=true;
    info "Availability confirmation sent successfully.";
  }
  else
  {
    info "Mail API did not confirm delivery. Check connection and retry.";
    info response.get("status");
  }
}
catch (error)
{
  info "Mail connection failed. Confirmation remains unsent; review the connection before
    retrying.";
}
```

Recipient and sender mapping

```
payload.put("fromAddress","caleb1510@zohomail.in");
payload.put("toAddress",request.Email_Address);
```

A failed or unsuccessful response does not set the confirmation flag. The function records diagnostic information for review.

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7. Email Content Design

Message section	Content
Subject	Request ID and availability / collection information.
Greeting	Dear Library Member,
Availability statement	The request has been approved and is available for collection.
Request-details table	Request ID, category, number of copies, preferred collection date, time and availability status.
Collection instructions	Bring the library ID, quote the request ID, and confirm the return due date with staff.
Footer	Library Book Issue Management identification.

8. Integration Test Results

Test	Observed result
Connection authorization	Authorized; application-specific connection access enabled.
Approved test recipient	caleb1510@zohomail.in, the owner mailbox.
Structured confirmation	Received message displayed a greeting, details table and collection instructions.
Confirmation tracking	Successful API response set Confirmation_Sent to true.
Mock recipient handling	example.com test request was skipped; log stated that no email was sent.
Repeat ordinary edit	No duplicate confirmation in the previously tested flow.

The documentation uses existing approved delivery evidence. No extra external test email was sent while preparing these PDFs.

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9 - Implementation Evidence

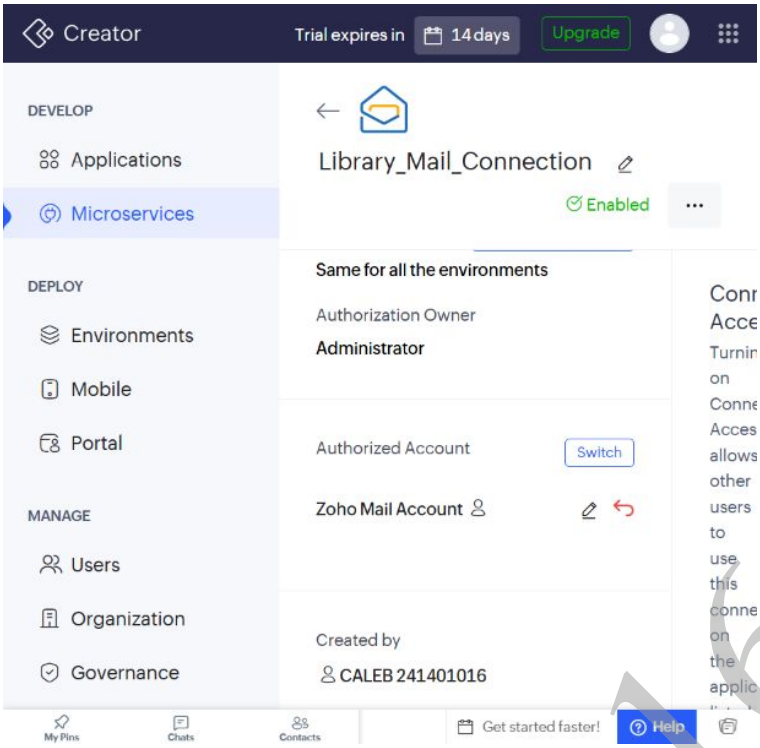


Figure 7.1. Zoho Mail connection evidence showing the authorized connection.

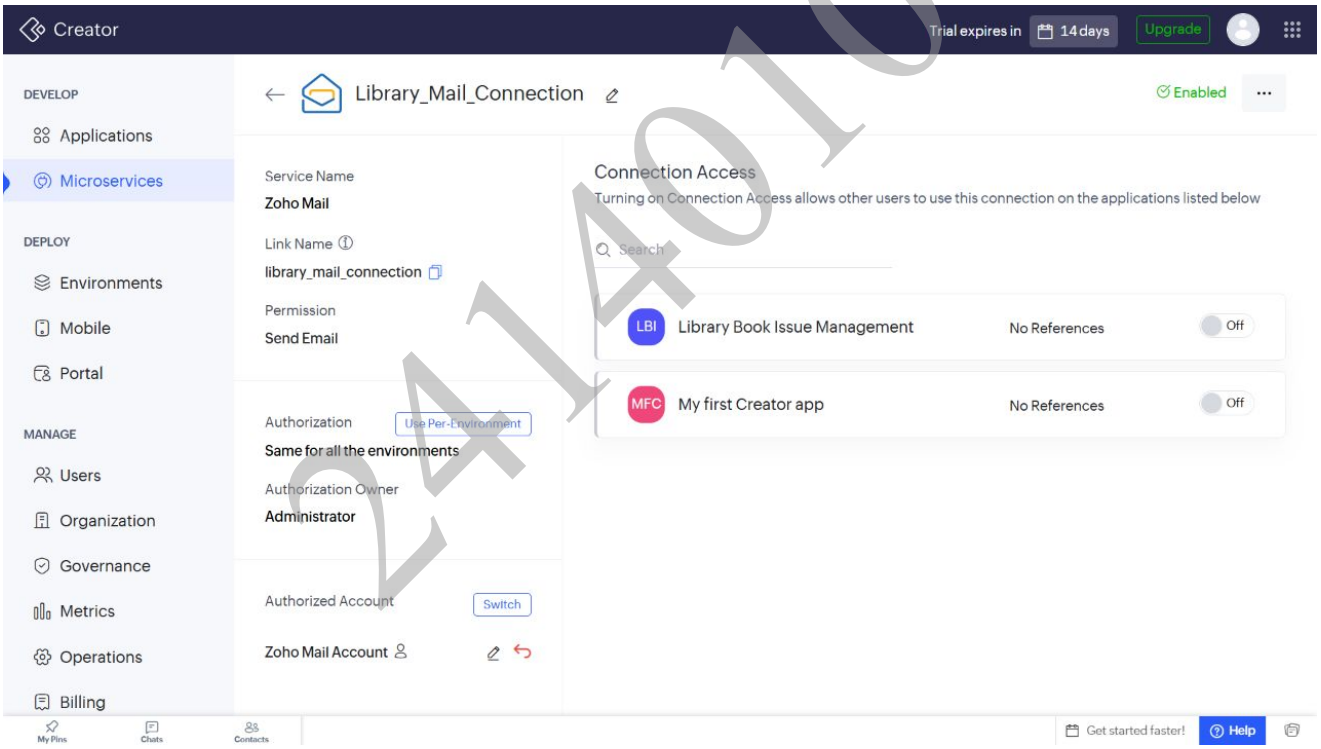


Figure 7.2. Application access for Library_Mail_Connection is enabled for the library app.

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9 - Implementation Evidence

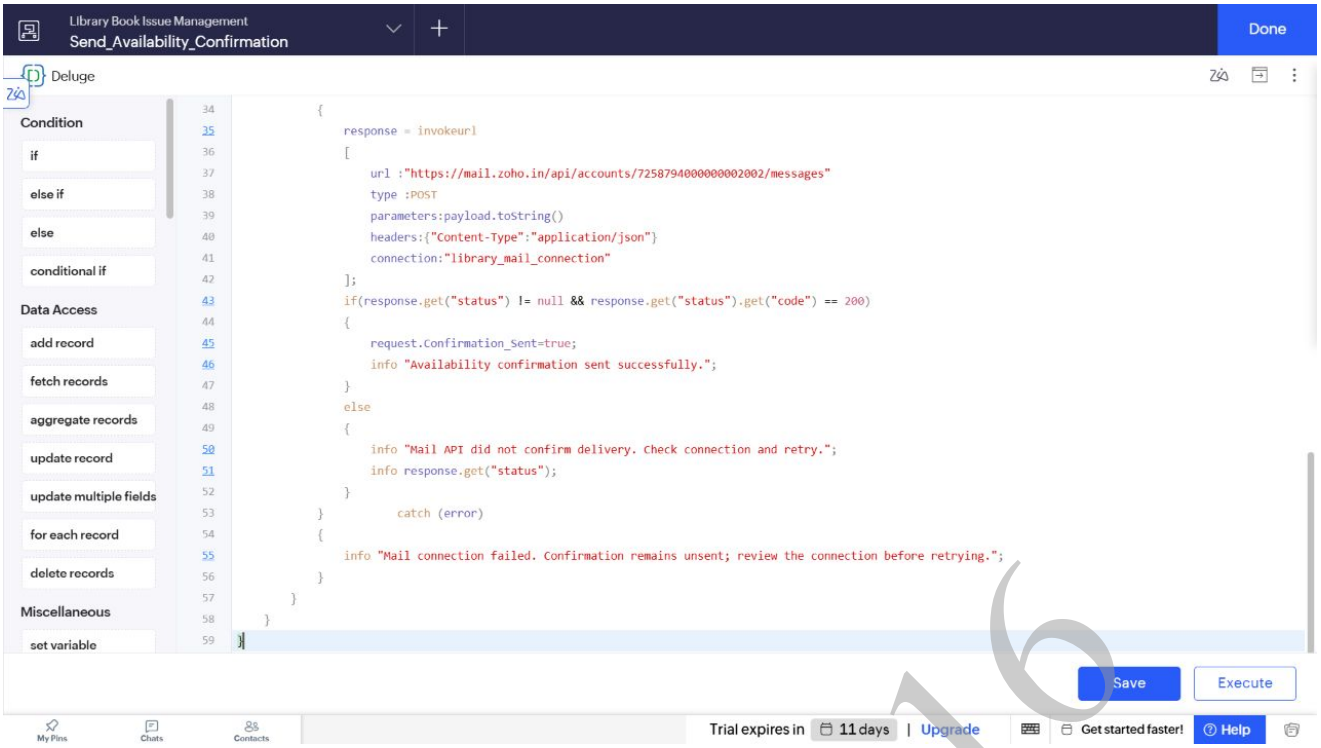


Figure 7.3. Saved function source showing POST to Zoho Mail, the named connection and success/error handling.

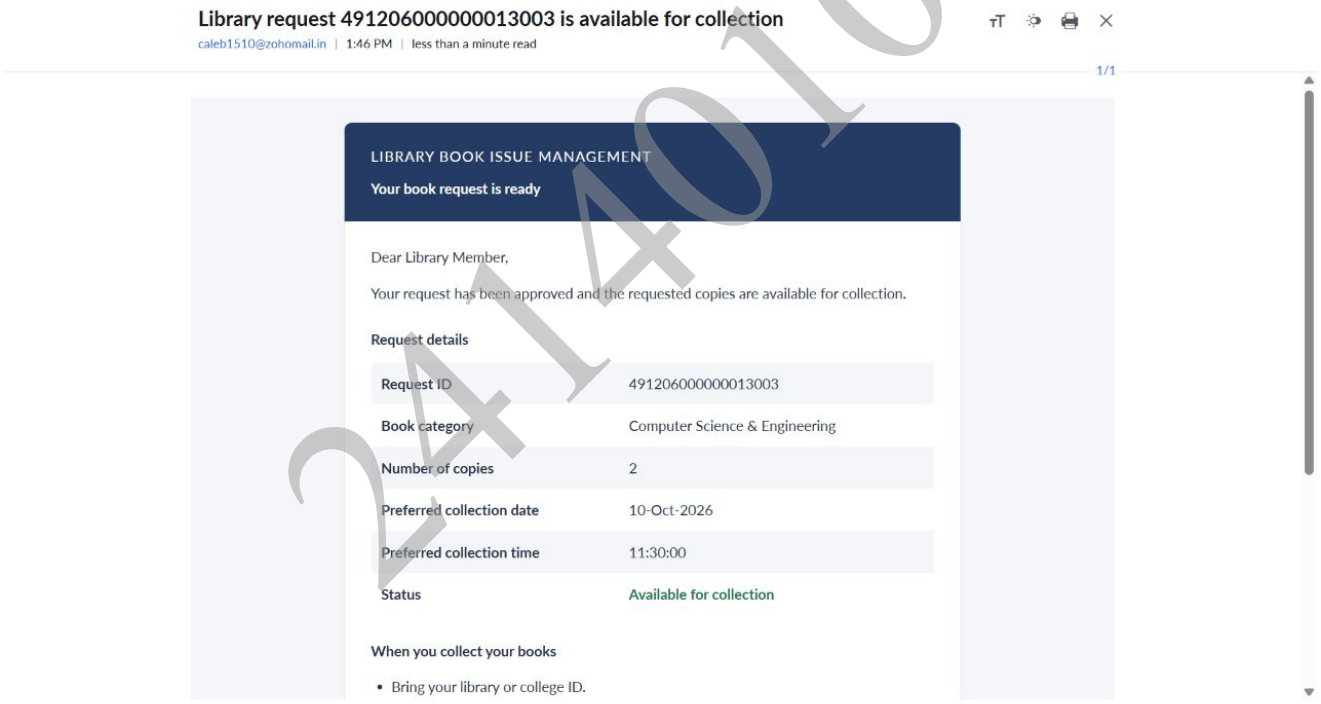


Figure 7.4. Actual delivered structured confirmation in the owner mailbox, verified on 09 October 2026.

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9 - Implementation Evidence

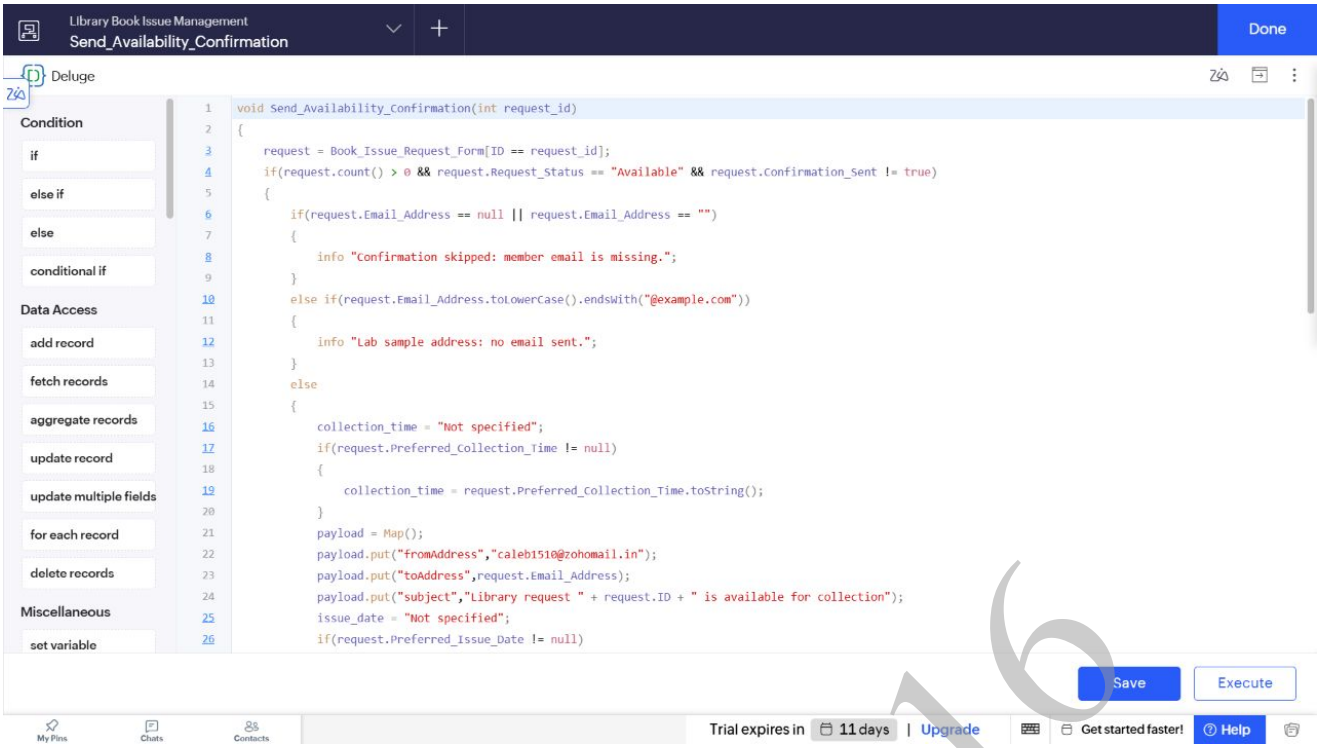


Figure 7.5. Saved function source: eligibility checks, example.com guard and the recipient mapped to Email_Address.

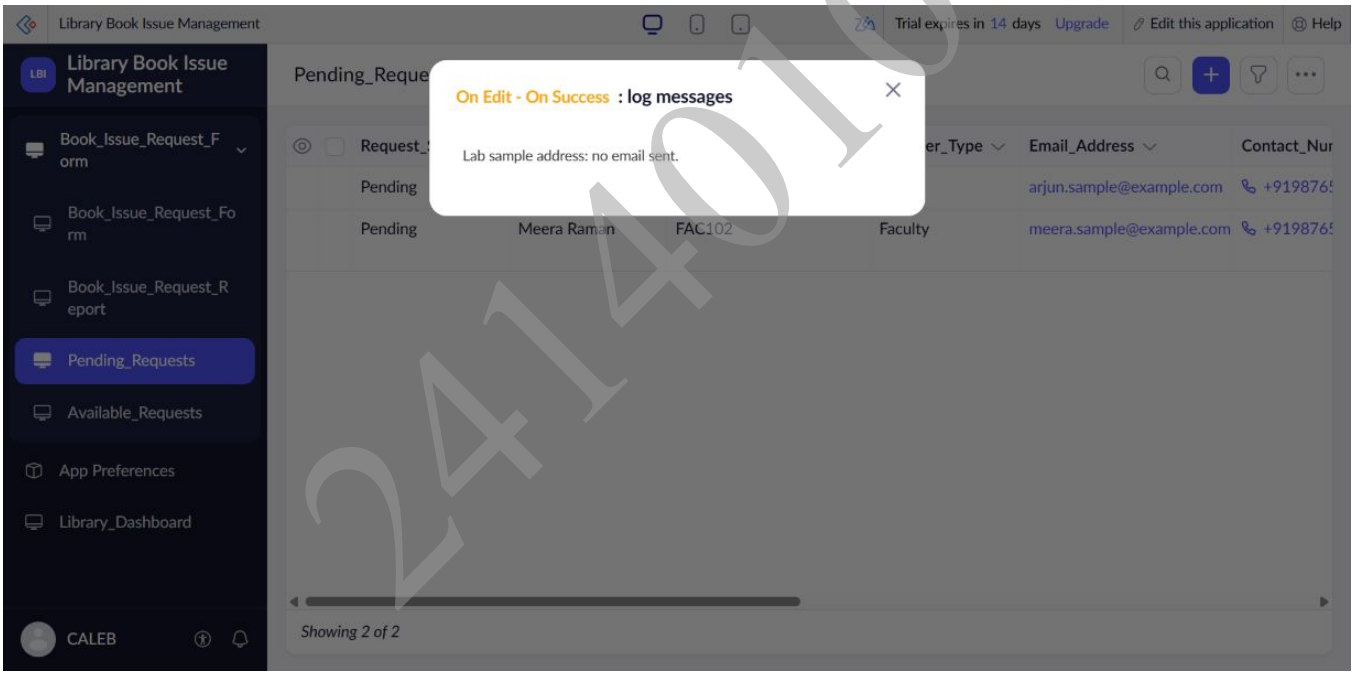


Figure 7.6. Earlier mock-address test: workflow log confirms that no email was sent to an example.com address.

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10. Expected Learning Outcomes

- Use an authorized Connection instead of embedding credentials.

- Connect a form workflow to a reusable API function.

- Build a clear HTML notification using request data.

- Distinguish API acceptance, received-email evidence and duplicate-send controls.

11. Result

The application was integrated with Zoho Mail using Library_Mail_Connection. An approved availability confirmation was delivered to the owner mailbox, and the structured HTML template was verified.

Conclusion

The integration informs a member when a request is available and records successful sending through Confirmation_Sent. The test demonstrates the configured Creator-to-Mail flow without sending notifications to fictitious addresses.

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